

PRD: SU to CG Trade Document Verification Agent

Part 2 · Builds on the Part 1 POC: same vision extraction, same store, same analytics layer. v1.0 · 2026-08-02

Problem

Every shipment's documents are validated by a CG team member who opens each attachment, reads every field, and checks it against what the customer requires — then types out what is wrong. Rules live in people's heads, so a new hire errs for weeks, nobody can see how many documents are pending, and there is no audit trail when a dispute surfaces. The result is 2 to 4 amendment cycles per shipment at 4 to 24 hours each, with CG bandwidth capping how fast shipments clear.

Personas

Meera — CG validator (3 yrs)

Checks ~40 documents a day across a dozen customers' unwritten rule sets. Judged on one thing: errors that reach the customer (a wrong HS code means a customs hold and a penalty). Cares about catching every discrepancy without re-reading every field, and never being the reason a shipment sat for a day.

Rahul — SU documentation executive

His job feels done when the doc-set email is sent. Judged on dispatch speed; every amendment email is unplanned rework. Cares about knowing exactly what to fix, in one pass — not "please recheck the invoice" ping-pong.

Jobs to be done

- (Meera)** When an SU document lands in my inbox, I want a field-by-field verdict against the customer's written requirements before I open the attachment, so that I spend my time only on the fields that are actually wrong.
- (Rahul)** When my documents have an issue, I want an amendment request that lists each field with what I sent and what was expected, so that I can fix everything in one cycle instead of three.

The flow (bold = human touchpoint)

SU emails the document -> agent detects it (trigger) -> agent extracts fields (Part 1 vision agent) -> agent compares against the customer rule set (deterministic) -> agent flags each field: match / mismatch / uncertain / missing -> agent drafts the reply -> **CG opens the verification result** -> **CG inspects flagged fields** (found vs expected, with the quoted evidence) -> **CG edits the draft if needed** -> **CG SENDS** -> **SU fixes and resends** (loop), or on a clean pass the docs go to the customer.

The three-party structure is untouched — SU sends, CG validates, the customer receives one clean set. The agent removes the reading and the typing, not the humans. It has no send capability at all.

North-star metric

Median turnaround: SU email arrival to CG reply sent (minutes). Computed from the system's own audit trail (`v_verifications.turnaround_minutes`), so a CG team lead can check it on Day 14 with one query against the manual baseline of hours. Guardrail so speed never buys errors: percentage of agent-approved documents later amended (target: 0).

Failure mode, and how it is stopped

Worst case: a false approval — the agent shows a wrong field as matched, Meera trusts the green tick, and a bad document reaches customs. Stopped four ways: **(1)** verdicts are deterministic rule checks against a written rule set, so no model judgment can reason a mismatch away; **(2)** any field below the confidence bar is *uncertain*, and uncertain blocks approval exactly like a mismatch — the agent never silently approves what it could not read; **(3)** every verdict carries the verbatim evidence quote, so CG can spot-check any field in seconds; **(4)** the agent cannot send — an approval only goes out through Meera's button, and the reply is rendered from the recorded check table, so it cannot claim anything the comparator did not find.

Out of scope this iteration, deliberately: real email integration (the trigger is a watched folder, per the brief), multi-customer rule packs, rule learning from amendment history, and cross-document consistency (invoice vs B/L). The last two are already named as Iteration 2 in the Part 1 PRD.